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MA20266: Statistical Inference

Group Project

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An Inference-Based Analysis of Gender Differences in Salary

Introduction

In today's data-driven world, analyzing real-world datasets is essential for understanding patterns, relationships, and potential disparities across different groups. This project focuses on examining a salary dataset to study the impact of gender on salary and other related factors using statistical methods.

The dataset used in this project was obtained from Kaggle, a widely used platform for data science and machine learning resources. Since real-world data often contains inconsistencies such as missing values, duplicates, and formatting issues, an important initial step in this project involved data cleaning. This process included handling missing data, removing duplicates, correcting data types, and ensuring consistency across all variables to make the dataset suitable for analysis.

After preprocessing, exploratory data analysis (EDA) was performed to understand the structure and key characteristics of the dataset. This step helped identify trends, distributions, and relationships between variables such as gender, salary, and job-related attributes.

To further analyze gender-based differences, three statistical tests were conducted:

- **Two - sample t-test** was used to compare the mean salaries of men and women, helping to identify whether any observed difference in average salary is statistically significant.
- **F-test** was performed to compare the variance of salaries between men and women, determining whether there is a significant difference in salary variability across genders.
- **Regression Analysis** was used to study the relationship between gender and multiple independent variables in the dataset. This helps in understanding how various factors collectively relate to gender.

Through this approach, the project aims to provide a comprehensive comparison of gender with respect to salary and other influencing factors. The findings of this analysis can offer insights into potential inequalities and contribute to a better understanding of fairness and distribution in the workplace.

Problem Statement

This project aims to analyze a salary dataset (sourced from Kaggle) to examine whether gender has a significant impact on salary and related factors. Using statistical methods such as

regression analysis, F-test, and two-sample t-test, the study investigates differences in salary distribution, variance, and relationships between gender and other variables.

Data Preprocessing: Before conducting the statistical analysis, the dataset was preprocessed to ensure consistency and suitability for inference. Observations containing null or missing values were removed. The data were checked for outliers, particularly in the salary variable. In addition, categorical variables such as gender, education level, and job title were transformed into dummy variables wherever required, so that they could be incorporated appropriately into the regression model.

Shape of dataset: (6688, 6)

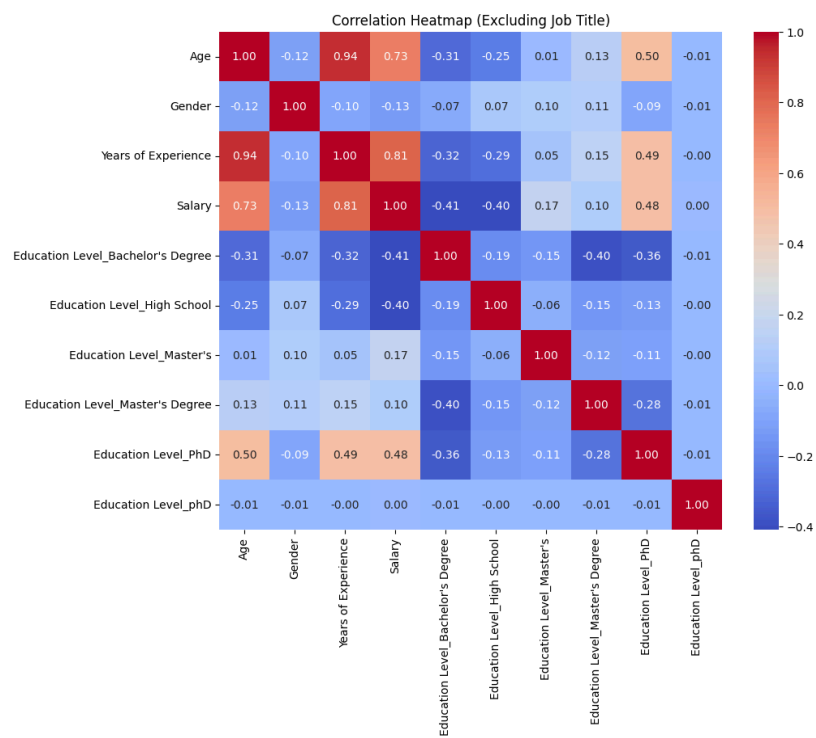
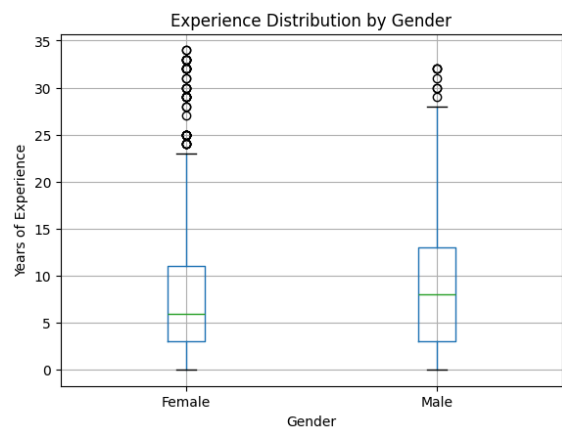
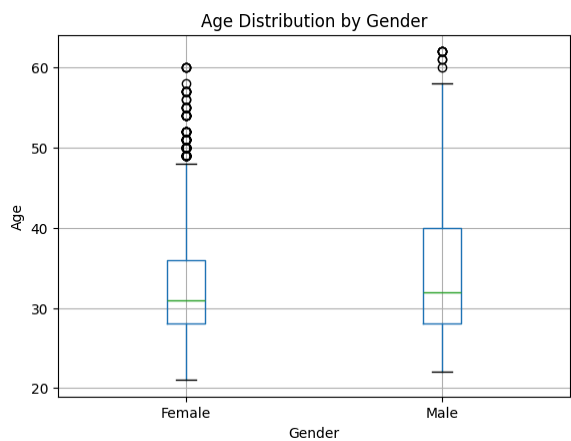
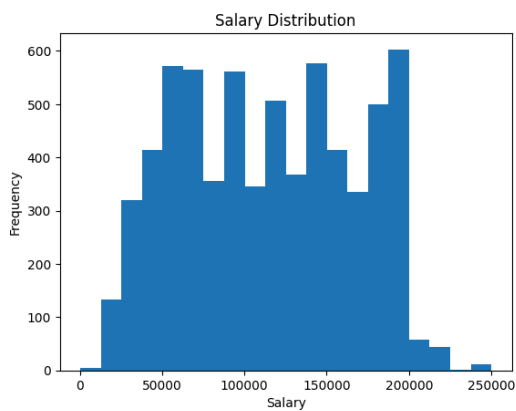
```
===== GENDER COUNT =====
Gender      count
Male        3671
Female      3013
Name: count, dtype: int64
```

```
===== SALARY SUMMARY BY GENDER =====
Gender      count      mean      median      std      min      max
Female      3013  107889.0  105000.0  52723.61  500.0  220000.0
Male        3671  121395.7  120000.0  52098.63  350.0  250000.0
```

```
===== SALARY SUMMARY BY EDUCATION LEVEL =====
Education Level      count      mean      median      std
PhD                  1369  165651.46  170000.0  34339.75
Master's Degree      1858  130078.38  130000.0  40649.72
Bachelor's Degree    3021   95082.91   80000.0  44091.79
High School           436   34415.61   30000.0  16563.41
```

```
===== SALARY SUMMARY BY YEARS OF EXPERIENCE =====
Years of Experience      count      mean      median
0.0                      120   29680.23   25000.0
0.5                       1   35000.00   35000.0
1.0                      540   46992.85   40000.0
1.5                       12   36279.17   37500.0
2.0                      606   58651.65   55000.0
3.0                      602   72944.41   65000.0
4.0                      522   83332.09   73916.5
5.0                      399  103111.09   95000.0
6.0                      438  111891.15  106193.5
7.0                      352  122171.04  120000.0
8.0                      423  126543.63  120000.0
9.0                      380  138021.46  135000.0
10.0                     192  131690.32  130000.0
11.0                     320  153060.32  160000.0
12.0                     294  153398.06  160000.0
```

Exploratory Data Analysis (EDA):



TWO SAMPLE T-TEST :

The purpose of this test is to determine whether the mean salary of male employees differ significantly from the mean salary of female employees in the given dataset. Since the salary variable is numerical and gender divides the observations into two independent groups, a two-sample t-test is an appropriate method for comparing the average salaries. We used Welch's two sample t-test which does not require the assumption that the population variances of the two groups are equal.

Hypothesis:

Let:

- μ_m = mean salary of male employees
- μ_f = mean salary of female employees

1. Two-Tailed Test: This version checks whether the two mean salaries are different, without specifying direction.

$$H_0 : \mu_m = \mu_f \quad H_1 : \mu_m \neq \mu_f$$

2. Right-Tailed Test: This version checks whether male employees have a higher mean salary than female employees.

$$H_0 : \mu_m \leq \mu_f \quad H_1 : \mu_m > \mu_f$$

3. Left-Tailed Test: This version checks whether male employees have a lower mean salary than female employees.

$$H_0 : \mu_m \geq \mu_f \quad H_1 : \mu_m < \mu_f$$

Since the main interest in this study is whether male salaries are higher than female salaries, the right-tailed test is more relevant.

The analysis for this problem has been carried out using programming (Python). The implementation code along with the corresponding outputs has been included below in the form of screenshots for reference and verification.

Method Implemented:

```
#Two-sample t-test
# Welch's t-test (does not assume equal variances)
t_stat, p_two_tailed = stats.ttest_ind(male_salary, female_salary, equal_var=False)

print("\n----- Welch Two-Sample t-Test -----")
print(f"t-statistic : {t_stat:.6f}")
print(f"Two-tailed p : {p_two_tailed:.6f}")

#One-tailed p-values
# H1: mean(Male) > mean(Female)
if t_stat > 0:
    p_right = p_two_tailed / 2
else:
    p_right = 1 - (p_two_tailed / 2)

# H1: mean(Male) < mean(Female)
if t_stat < 0:
    p_left = p_two_tailed / 2
else:
    p_left = 1 - (p_two_tailed / 2)

print("\n----- One-Tailed p-values -----")
print(f"Right-tailed p-value [H1: mu_male > mu_female] : {p_right:.6f}")
print(f"Left-tailed p-value [H1: mu_male < mu_female] : {p_left:.6f}")

#95% confidence interval for (mu_male - mu_female)
mean_m = male_salary.mean()
mean_f = female_salary.mean()
var_m = male_salary.var(ddof=1)
var_f = female_salary.var(ddof=1)
n_m = len(male_salary)
n_f = len(female_salary)

alpha = 0.05
t_critical = stats.t.ppf(1 - alpha / 2, df_welch)

ci_low = mean_diff - t_critical * se
ci_high = mean_diff + t_critical * se

print("\n----- 95% Confidence Interval for (mu_male - mu_female) -----")
print(f"Mean difference : {mean_diff:.2f}")
print(f"95% CI : ({ci_low:.2f}, {ci_high:.2f})")

#Decision at 5% significance level
print("\n----- Decision at alpha = 0.05 -----")

# Two-tailed test
if p_two_tailed < 0.05:
    print("Two-tailed test: Reject H0")
else:
    print("Two-tailed test: Fail to reject H0")

# Right-tailed test: H1 mu_male > mu_female
if p_right < 0.05:
    print("Right-tailed test: Reject H0")
else:
    print("Right-tailed test: Fail to reject H0")

# Left-tailed test: H1 mu_male < mu_female
if p_left < 0.05:
    print("Left-tailed test: Reject H0")
else:
    print("Left-tailed test: Fail to reject H0")
```

Result:

```
----- Welch Two-Sample t-Test -----
t-statistic : 10.473640
Two-tailed p : 0.000000

----- One-Tailed p-values -----
Right-tailed p-value [H1: mu_male > mu_female] : 0.000000
Left-tailed p-value [H1: mu_male < mu_female] : 1.000000

----- 95% Confidence Interval for (mu_male - mu_female) -----
Mean difference : 13500.87
95% CI : (10973.94, 16027.81)

----- Decision at alpha = 0.05 -----
Two-tailed test: Reject H0
Right-tailed test: Reject H0
Left-tailed test: Fail to reject H0
```

Decision Rule:

The level of significance used is: $\alpha=0.05$

1. **Two-Tailed Test:** Since the two-tailed p-value is less than 0.05, we reject the null hypothesis.
2. **Right-Tailed Test:** Since the right-tailed p-value is less than 0.05, we reject the null hypothesis

3. **Left-Tailed Test:** Since the left-tailed p-value is greater than 0.05, we fail to reject the null hypothesis.

F-TEST FOR VARIANCE :

F-test is used to compare variability (variance) of two populations.

Here we are taking the first population as Male Salaries and the second population as Female Salaries.

Test Statistic:

$$F = S_1^2 / S_2^2$$

Where S_1^2, S_2^2 are sample variances of population.

Hypothesis :

Right-Tailed Test : This test checks whether the variance of salaries of male employees is greater than that of female employees.

$$H_0 : \sigma_m^2 \leq \sigma_f^2$$
$$H_1 : \sigma_m^2 > \sigma_f^2$$

Left-Tailed Test : This test checks whether the variance of salaries of male employees is less than that of female employees.

$$H_0 : \sigma_m^2 \geq \sigma_f^2$$
$$H_1 : \sigma_m^2 < \sigma_f^2$$

Two-Tailed Test : This test checks whether the variances of salaries of male and female employees are different, without specifying any direction.

$$H_0 : \sigma_m^2 = \sigma_f^2$$
$$H_1 : \sigma_m^2 \neq \sigma_f^2$$

Method Implemented:

```
#Load dataset
df = pd.read_excel("/kaggle/input/datasets/jahnavireddykallu/salary/Salary_data.xlsx.xlsx")
# Clean (important)
df["Gender"] = df["Gender"].str.strip()
# Extract data
male = df[df["Gender"] == "Male"]["Salary"].dropna().values
female = df[df["Gender"] == "Female"]["Salary"].dropna().values
#Variances
var_m = np.var(male, ddof=1)
var_f = np.var(female, ddof=1)
print("Male variance:", var_m)
print("Female variance:", var_f)
```

```
# Left-tailed test
F = var_m / var_f
df1 = len(male) - 1
df2 = len(female) - 1
alpha = 0.05
F_critical = f.ppf(alpha, df1, df2)
p_value = f.cdf(F, df1, df2)
print("F:", F)
print("F-critical:", F_critical)
print("p-value:", p_value)
if F < F_critical:
    print("Reject H0: Male variance is smaller")
else:
    print("Fail to reject H0")
```

```
# Right-tailed test
F = var_m / var_f
df1 = len(male) - 1
df2 = len(female) - 1
alpha = 0.05
F_critical = f.ppf(1 - alpha, df1, df2)
p_value = 1 - f.cdf(F, df1, df2)
print("F:", F)
print("F-critical:", F_critical)
print("p-value:", p_value)
if F > F_critical:
    print("Reject H0: Male variance is greater")
else:
    print("Fail to reject H0")
```

```
# Two-tailed test
# Always put larger variance on top
if var_m > var_f:
    F = var_m / var_f
    df1 = len(male) - 1
    df2 = len(female) - 1
else:
    F = var_f / var_m
    df1 = len(female) - 1
    df2 = len(male) - 1
alpha = 0.05
F_critical = f.ppf(1 - alpha/2, df1, df2)
p_value = 2 * (1 - f.cdf(F, df1, df2))
print("F:", F)
print("F-critical:", F_critical)
print("p-value:", p_value)
if F > F_critical:
    print("Reject H0")
else:
    print("Fail to reject H0")
```

Result:

```
Male variance: 2713652140.430622
Female variance: 2779779055.1600246
-----Right-tailed test-----
F: 0.9762114493931979
F-critical: 1.0589631161934838
p-value: 0.7561062903919575
Fail to reject H0
-----Left-tailed test-----
F: 0.9762114493931979
F-critical: 0.9444965709005915
p-value: 0.2438937096080424
Fail to reject H0
-----Two-tailed test-----
F: 1.0243682356128774
F-critical: 1.0704081011758682
p-value: 0.4877874192160849
Fail to reject H0
```


Decision Rule :

The level of Significance used is $\alpha = 0.05$

1. Right Tailed Test:

Reject H_0 if :

$$F > F_{\alpha/2}(df1, df2)$$

Otherwise :

Fail to reject H_0

2. Left Tailed Test:

Reject H_0 if :

$$F < F_{\alpha/2}(df1, df2)$$

Otherwise :

Fail to reject H_0

3. Two Tailed Test:

Reject H_0 if :

$$F > F_{\alpha/2}(df1, df2) \text{ or } F > F_{1-\alpha/2}(df1, df2)$$

Otherwise :

Fail to reject H_0

REGRESSION TEST :

In regression , we estimate parameters using least square method and statistical inference involving hypothesis testing and likelihood-based approaches.

We consider the standard regression model

$$Y_{nx1} = X_{Nxp} \beta_{px1} + \varepsilon_{nx1} \quad \text{where } Y \text{ is Dependent vector of size } n \times 1, X \text{ is Design matrix of size } nxp, \\ \varepsilon \text{ is the constant in regression}$$

Hypothesis:

Null Hypothesis (H_0):

$$\beta_{male} = 0$$

Implies that Gender has no effect on salary (no wage gap after controlling for other factors)

Alternative Hypothesis (H_1):

$$\beta_{male} \neq 0$$

Gender has a significant effect on salary

Interpretation of Outcome

- If we reject $H_0 \Rightarrow$ evidence of a gender salary gap (positive or negative depending on sign of coefficient)
- If we fail to reject $H_0 \Rightarrow$ no statistically significant gender effect after controlling for education and job title

Method Implemented

We consider the multi linear regression model:

$$Salary_i = \beta_0 + \beta_1 Male_i + \beta_2 X_{2i} + \dots + \beta_k X_{ki} + \varepsilon_i$$

where:

- $Salary_i$ is the dependent variable (employee salary),
- $Male_i$ is a binary variable for gender,
- $X_{2i}, \dots, X_{ki}$ represent dummy variables for education level and job title,
- ε_i is the error term.

We estimate the parameter vector β using Ordinary Least Squares (OLS).

```
import pandas as pd
import statsmodels.api as sm
data=pd.read_excel("Salary_data.xlsx")

#converting the categorical variables to numeric
data["Male"] = (data["Gender"] == "Male").astype(int)
data = pd.get_dummies(data, columns=["Education Level", "Job Title"], drop_first=True)
data = data.drop(columns=["Gender"])
data = data.dropna() # dropping null values

X = data.drop(columns=["Salary"]) # independent variables
X = X.astype(float)
y = data["Salary"] # dependent variable
X = sm.add_constant(X) # adds intercept
model = sm.OLS(y, X).fit()
print(model.summary())
```

Result

OLS Regression Results						
=====						
Dep. Variable:	Salary	R-squared:	0.889			
Model:	OLS	Adj. R-squared:	0.885			
Method:	Least Squares	F-statistic:	259.2			
Date:	Tue, 14 Apr 2026	Prob (F-statistic):	0.00			
Time:	17:18:30	Log-Likelihood:	-74838.			
No. Observations:	6685	AIC:	1.501e+05			
Df Residuals:	6484	BIC:	1.514e+05			
Df Model:	200					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	1.983e+04	1.81e+04	1.097	0.273	-1.56e+04	5.53e+04
Age	1039.2915	100.190	10.373	0.000	842.885	1235.698
Years of Experience	4084.8458	124.246	32.877	0.000	3841.283	4328.409
Male	1487.8119	511.760	2.907	0.004	484.593	2491.031
Education Level_Bachelor's Degree	-5.147e+04	1085.588	-47.412	0.000	-5.36e+04	-4.93e+04
Education Level_High School	-6.057e+04	2039.567	-29.700	0.000	-6.46e+04	-5.66e+04
Education Level_Master's	-5014.6347	1582.509	-3.169	0.002	-8116.874	-1912.396
Education Level_Master's Degree	-4.258e+04	1171.245	-36.356	0.000	-4.49e+04	-4.03e+04
Education Level_PhD	-3.597e+04	1398.759	-25.716	0.000	-3.87e+04	-3.32e+04
Education Level_phD	-1.004e+04	1.8e+04	-0.559	0.576	-4.53e+04	2.52e+04

For Male variable:

- Coefficient = 1487.81
- t-statistic = 2.907
- p-value = 0.004

Since p-value is less than 0.005, we reject H_0

Practical Implications

1. Two-Sample t-Test:

The results of the two-sample t-test indicate that there is a statistically significant difference in the mean salaries of male and female employees. This suggests the presence of a gender-based pay gap in the dataset. In practical terms, this highlights a potential issue of pay inequality, indicating that organizations may need to review compensation policies to ensure fairness and equal pay practices.

2. F-Test:

The F-test shows that there is no significant difference in the variance of salaries between male and female employees. This implies that although the average salaries

differ, the spread or dispersion of salaries within both groups is similar. Practically, this means that the observed disparity is not due to differences in salary variability, but rather due to a consistent difference in average pay levels.

3. **Regression Analysis:**

The regression results show a high explanatory power ($R^2 = 0.889$), indicating that a large proportion of the variation in salary is explained by the model. Variables such as age and years of experience have strong positive and statistically significant effects on salary. The regression results show a high explanatory power ($R^2 = 0.889$), indicating that a large proportion of the variation in salary is explained by the model. Variables such as age and years of experience have strong positive and statistically significant effects on salary. Importantly, the coefficient for the male variable is positive (1487.81) and statistically significant (p-value = 0.004). This indicates that, holding other factors constant, male employees earn on average approximately 1488 units more than female employees.